Area, Surface Area & Volume

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 24.5. The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(3)$.

Final Answer: 23. Substitute $x = 3$: $f(3) = 6(3) + 5 = 23$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 85°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (50^\circ + 45^\circ) = 85^\circ$.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A right circular cylinder has radius 3 cm and height 8 cm. What is its total surface area in terms of π?

The total surface area includes the two circular bases and the lateral curved surface:

$\text{TSA} = 2\pi r^2 + 2\pi rh = 2\pi(3^2) + 2\pi(3)(8) = 18\pi + 48\pi = \mathbf{66\pi}$ cm^2

5 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A cuboid has a length of USD 8$cm, a width of USD 5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

The volume of a cuboid is calculated using the formula: Volume = length × width × height. Given: length = USD 8$cm, width = USD 5$ cm, height = USD 3$cm. Volume = 8 \text{ cm} \times 5 \text{ cm} \times 3 \text{ cm} Volume = 40 \text{ cm}^2 \times 3 \text{ cm} Volume = 120 \text{ cm}^3$. $Therefore, the volume of the cuboid is USD 120$ cm^3.

6 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Find the area of a triangle with base 10 and height 6.

Apply the triangle area formula $A = \frac{1}{2}bh$: $A = \frac{1}{2} \times 10 \times 6 = 5 \times 6 = \mathbf{30}$ square units Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the area of a circle with diameter 10.

Step 1: Determine the radius: Radius is half the diameter: $r = \frac{d}{2} = \frac{10}{2} = 5$ Step 2: Apply the circle area formula: $A = \pi r^2 = \pi(5^2) = \mathbf{25\pi} \approx 78.54$ The exact area is $\mathbf{25\pi}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circle has a radius of 12 cm. What is the area of a sector formed by a central angle of 75°?

The formula for the area of a circle sector is: $A = \frac{\theta}{360^\circ}\pi r^2 = \frac{75}{360}\pi(12^2) = \frac{5}{24} \cdot 144\pi = 5 \cdot 6\pi = \mathbf{30\pi}$ cm^2

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the volume of a rectangular prism $4 \times 3 \times 5$.

Step 1: Apply rectangular prism volume formula: $V = l \times w \times h$ Step 2: Compute product: $V = 4 \times 3 \times 5 = 12 \times 5 = \mathbf{60}$ The volume is 60 cubic units.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two similar cylinders have radii 4 cm and 6 cm. If the volume of the smaller is 160π cm³, find the volume of the larger.

$k = 6/4 = 1.5$. Volume ratio $= 1.5^3 = 3.375$. Volume $= 160\pi \times 3.375 = 540\pi$ cm³.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedA school administrator randomly surveys 120 high school seniors at Oakwood High School. In the sample, 65% report planning to attend a four-year college, with a margin of error of $\pm 4\%$. Which of the following conclusions is most statistically sound? (A) Exactly 65% of all Oakwood High seniors will attend college. (B) It is plausible that between 61% and 69% of all Oakwood High seniors plan to attend a four-year college. (C) Between 61% and 69% of all high school seniors in the state plan to attend college. (D) To reduce the margin of error by half, the administrator should survey 240 seniors.

Correct Answer: (B). The margin of error creates a plausible confidence interval $65\% \pm 4\% = [61\%, 69\%]$. Because the random sample was drawn exclusively from Oakwood High seniors, the conclusion generalizes only to that school's senior class (eliminating C). To halve the margin of error, sample size must quadruple ($4 \times 120 = 480$), making D incorrect.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · AdvancedUsing the student prep table (Studied: 54 passed, 6 failed; Did not study: 18 passed, 22 failed; Total passed: 72 out of 100), are the events 'passed' and 'studied' independent?

Step 1: Calculate marginal probability: $P(\text{Passed}) = \frac{72}{100} = 0.72$. Step 2: Compare with conditional probability: $P(\text{Passed} \mid \text{Studied}) = \frac{54}{60} = 0.90$. Since $P(\text{Passed} \mid \text{Studied}) = 0.90 eq 0.72 = P(\text{Passed})$, the events are not independent (studying is positively associated with passing).

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A line passes through $(1, 2)$ with slope 3. Find y when $x = 4$.

Method 1 (Point-Slope Form): $y - y_1 = m(x - x_1) \Rightarrow y - 2 = 3(x - 1)$ Substitute $x = 4$: $y - 2 = 3(4 - 1) = 3(3) = 9 \Rightarrow y = 9 + 2 = \mathbf{11}$ Method 2 (Rate of Change / Delta Method): $\Delta x = 4 - 1 = 3$. Since slope $m = \frac{\Delta y}{\Delta x} = 3$, $\Delta y = m \cdot \Delta x = 3 \cdot 3 = 9$. New $y = 2 + 9 = \mathbf{11}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line has slope 3 and passes through $(0, 2)$. Find y when $x = 2$.

Step 1: Identify the line's equation. Since $(0, 2)$ is the y-intercept, $b = 2$. Given slope $m = 3$: $y = mx + b \Rightarrow y = 3x + 2$ Step 2: Evaluate at $x = 2$: $y = 3(2) + 2 = 6 + 2 = \mathbf{8}$ Alternative Delta Method: For every 1 unit increase in x, y increases by 3. From $x = 0$ to $x = 2$ is $+2$ units, so $y = 2 + 2(3) = \mathbf{8}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Lines & Angles

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

Identify coefficients: $a = 2$, $b = 5$, $c = -4$. $x = \frac{-5 \pm \sqrt{5^2 - 4(2)(-4)}}{2(2)} = \frac{-5 \pm \sqrt{25 + 32}}{4} = \frac{-5 \pm \sqrt{57}}{4}$ $x_1 = \frac{-5 + 7.5498}{4} = \frac{2.5498}{4} \approx 0.64$ $x_2 = \frac{-5 - 7.5498}{4} = \frac{-12.5498}{4} \approx -3.14$ Solutions: $x = 0.64$ or $x = -3.14$ (both to 2 d.p.).

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Triangles & Congruence

Practice 3.2: What is the domain of the function $f(x) = \frac{\sqrt{x+5}}{x-3}$?

For the radical in the numerator to be defined, the radicand must be non-negative: $x + 5 \geq 0 \Rightarrow x \geq -5$. For the rational expression to be defined, the denominator cannot equal zero: $x - 3 eq 0 \Rightarrow x eq 3$. Combining both restrictions gives $x \geq -5$ and $x eq 3$, or in interval notation $[-5, 3) \cup (3, \infty)$.

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17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning**

For $f(x) = x^2$, find the average rate of change on $[1, 3]$.

The average rate of change on the closed interval $[a, b]$ is the slope of the secant line: Average Rate of Change $= \frac{f(b) - f(a)}{b - a}$ Compute function values at endpoints: $f(1) = 1^2 = 1, \quad f(3) = 3^2 = 9$ Substitute into the slope formula: $\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = \frac{8}{2} = \mathbf{4}$

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Right Triangles & Trigonometry**

Level 3 · AdvancedA quadratic function $f(x) = ax^2 + bx + c$ passes through the points $(-2, 11)$, $(0, 3)$, and $(2, 7)$. What is the value of $a + b + c$?

Solution via Desmos Table Regression:Click the + button and add a Table. Enter $x_1 = \{-2, 0, 2\}$ and $y_1 = \{11, 3, 7\}$.In the next expression line, type the regression template: y1 ~ a*x1^2 + b*x1 + c.Desmos instantly calculates the parameters: $a = 1$, $b = -1$, and $c = 3$.Calculate the target value: $a + b + c = 1 + (-1) + 3 = 3$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Polygons & Quadrilaterals**

A population of bacteria in a petri dish starts with 250 organisms and quadruples every 5 hours. How many hours will it take for the population to reach 16,000?

Step 1: Set up the exponential growth equation: $P(t) = 250 \times 4^{t/5} = 16000.$ Step 2: Divide both sides by 250: $4^{t/5} = \frac{16000}{250} = \frac{1600}{25} = 64.$ Step 3: Express 64 as a power of 4: $64 = 4^3 \Rightarrow 4^{t/5} = 4^3 \Rightarrow \frac{t}{5} = 3 \Rightarrow t = 15$ hours. It will take exactly 15 hours.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Circles**

How many real solutions does $x^2 + 2x + 5 = 0$ have?

Use the discriminant $\Delta = b^2 - 4ac$ from the quadratic formula: For $x^2 + 2x + 5 = 0$, $a = 1$, $b = 2$, $c = 5$: $\Delta = (2)^2 - 4(1)(5) = 4 - 20 = -16$ Since $\Delta < 0$, the equation has no real solutions (it has 2 complex conjugate solutions, $-1 \pm 2i$). In the xy-plane, the parabola never crosses the x-axis.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 45^\circ) = 88^\circ$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 87° ✓

b. 97°

c. 79°

d. 132°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (48^\circ + 45^\circ) = 87^\circ$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70° ✓

b. 80°

c. 62°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 59^\circ) = 70^\circ$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A sphere has radius 8. Find its surface area in terms of π.

a. 128π

b. 262π

c. 256π ✓

d. 64π

Surface area $= 4\pi r^2 = 4\pi(8)^2 = 256\pi$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A cylinder has radius 14 and height 3. Find its volume in terms of π.

a. 588π ✓

b. 42π

c. 1176π

d. 598π

Volume $= \pi r^2 h = \pi(14)^2(3) = 588\pi$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Volume of a cylinder with radius 2 and height 5 (in terms of π):

a. 10π

b. 25π

c. 40π

d. 20π ✓

$V = \pi r^2 h = \pi \times 2^2 \times 5 = 20\pi$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A rectangular prism has length 2, width 7, and height 9. What is its volume?

a. **126** ✓

b. 136

c. 18

d. 190

Volume $= lwh = 2 \times 7 \times 9 = 126$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Area of a circle with radius 3 (in terms of π):

a. 12π

b. 7π

c. 18π

d. 9π ✓

Applying Area, Surface Area & Volume principles, $A = \pi r^2 = \pi \times 3^2 = 9\pi$. Therefore, the verified correct option is 9π.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Surface area of a cube with side length 2:

a. 8

b. 16

c. 20

d. **24** ✓

Applying Area, Surface Area & Volume principles, $SA = 6s^2 = 6 \times 2^2 = 24$. Therefore, the verified correct option is 24.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Area of a rectangle with length 3 and width 5:

a. 18

b. 8

c. 16

d. **15** ✓

Applying Area, Surface Area & Volume principles, $A = lw = 3 \times 5 = 15$. Therefore, the verified correct option is 15.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two similar solids have corresponding side lengths in ratio $2:3$. Find the ratio of their volumes.

a. 4:9

b. **8:27** ✓

c. 6:9

d. 2:3

Volume scales with the cube of the linear ratio: $2^3 : 3^3 = 8:27$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Volume of a sphere with radius 3 (in terms of π):

a. 72π

b. 36π ✓

c. 18π

d. 47π

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times 3^3 = 36\pi$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A cone has radius 3 and height 4. Find the slant height.

a. 7

b. **5 ✓**

c. 8

d. 6

$\ell = \sqrt{r^2 + h^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Area of a triangle with base 8 and height 5:

a. 13

b. 40

c. 22

d. **20 ✓**

$A = \frac{1}{2}bh = \frac{1}{2} \times 8 \times 5 = 20.$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Surface area of a sphere with radius 3 (in terms of π):

a. 72π

b. 49π

c. 36π ✓

d. 18π

$SA = 4\pi r^2 = 4\pi \times 3^2 = 36\pi.$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two similar spherical storage tanks: Tank B has a surface area 9 times that of Tank A. If Tank A holds 250 liters, how many liters does Tank B hold?

a. 2,250 liters

b. **6,750 liters ✓**

c. 1,500 liters

d. 5,625 liters

Surface area scales by $k^2 = 9 \Rightarrow k = \sqrt{9} = 3$. Volume (capacity) scales by $k^3 = 3^3 = 27$. Therefore, Tank B holds $250 \times 27 = 6,750$ liters.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A topographical map has a scale of $1:25{,}000$. A regional forest reserve has an area of $8\ \text{cm}^2$ on the map. What is the actual ground area of the reserve in square kilometers (km^2)?

a. 0.5 km² ✓

b. 2.0 km²

c. 5.0 km²

d. 0.2 km²

On the map, $1\ \text{cm} = 25{,}000\ \text{cm} = 250\ \text{m} = 0.25\ \text{km}$. For area, $1\ \text{cm}^2 = (0.25\ \text{km})^2 = 0.0625\ \text{km}^2$. Therefore, an area of $8\ \text{cm}^2$ on the map represents $8 \times 0.0625 = 0.5\ \text{km}^2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

If each dimension of a rectangle is increased by 40%, by what percentage does the area of the rectangle increase?

a. 96% ✓

b. 40%

c. 80%

d. 140%

Increasing linear dimensions by 40% corresponds to a linear scale factor of $k = 1.40$. The new area is $k^2 = (1.40)^2 = 1.96$ times the original area. A factor of 1.96 represents a $(1.96 - 1) \times 100\% = 96\%$ increase.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A rectangle with length 4 and width 13 has a semicircle of diameter 13 attached to one of the short sides. What is the total area of the composite figure, rounded to two decimal places?

a. 118.37 ✓

b. 123.37

c. 184.73

d. 52

Rectangle area $= 4 \times 13 = 52$. Semicircle radius $= 6.5$, area $= \frac{1}{2}\pi r^2 \approx 66.37$. Total ≈ 118.37.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Circles

What is the surface area of a cube with edge length 4 cm?

a. 64 cm²

b. **96 cm² ✓**

c. 128 cm²

d. 144 cm²

A cube has 6 congruent square faces: $SA = 6s^2 = 6(4^2) = 6(16) = 96$ cm^2.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

A trapezoid has area 48 and height 6. Find the sum of its two parallel sides.

a. 8

b. 6

c. 18

d. **16 ✓**

$A = \frac{1}{2}(b_1 + b_2)h \Rightarrow b_1 + b_2 = \frac{2(48)}{6} = 16$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Foundations & Reasoning

A kite has diagonals 9 and 11. Find its area.

a. 40

b. $\frac{105}{2}$

c. $\frac{99}{2}$ ✓

d. 99

$A = \frac{1}{2}d_1d_2 = \frac{1}{2}(9)(11) = \frac{99}{2}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Coordinate Geometry

Area of triangle base 11 height 4:

a. 15

b. **22 ✓**

c. 44

d. 25

$A = \frac{1}{2}bh = \frac{1}{2} \times 11 \times 4 = 22$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability

A rhombus has diagonals 10 and 12. Find its area.

a. 22

b. **60** ✓

c. 62

d. 120

$A = \frac{1}{2}d_1d_2 = \frac{1}{2}(10)(12) = 60$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Lines & Angles

Triangle with sides 13, 14, 15. Find its area using Heron's formula.

a. 86

b. 81

c. **84** ✓

d. 91

$s = 21$, $A = \sqrt{s(s-a)(s-b)(s-c)} = 84$.

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